PowerHub PMS I2C Protocol

V1 (FW Version)
2025/11/11

REG MAP (Addr:0X50)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Power Enable Reg	0x00 W/R	LED Power Enable (0x00)	USB Power Enable (0x01)	Grove Red Power Enable (0x02)	Grove Blue Power Enable (0x03)	PWR485&PWRCAN Power Enable (0x04)	VAMeter Power Enable (0x05)	Charge Power Enable (0x06)										LED Power Enable: 1: LED power on 0: LED power off(default) USB Power Enable: 1: USB power on 0: USB power off(default) Grove Red Power Enable: 1: Grove red power on 0: Grove red power off(default) Grove Blue Power Enable: 1: Grove blue power on 0: Grove blue power off(default) PWR485&PWRCAN Power Enable: 1: PWR485&PWRCAN power on 0: PWR485&PWRCAN power off(default) VAMeter Power Enable: 1: VAMeter power on 0: VAMeter power off(default) Charge Power Enable: 1: Charge power on 0: Charge power off(default)
USB Mode	0x10 W/R	USB Mode (0x10)																USB Mode: 0: Slave mode, USB->ESP32(default) 1: Host mode(USB TypeA), ESP32->USB TypeA device 2: Host mode(USB TypeC), ESP32->USB TypeC device
PWR485&PWRCAN Config	0x20 W/R	Voltage LSB (0x20)	Voltage MSB (0x21)	Current Limit Value (0x22)	Power Output Control (0x23)	Power Direction Control (0x24)												Voltage: Range 3000mV~20000mV, step 20mv Current Limit Value: 0~232, Iout limit(mA) = (Current Limit Value+1)*8/0.62 Power Output Control: 1: Power output enable(default) 0: Power output disable Power Direction Control: 1: Power output direction 0: Power input direction(default)
SC8721 Config Read	0x90 R	Get SC8721 Config (0x90)	CSO (0x91)	SLOPE COMP (0x92)	VOUT SET_MSB (0x93)	VOUT SET_LSB (0x94)	GLOBAL CTRL (0x95)	SYS_SET (0x96)	RESERVED (0x97)	FREQ_SET (0x98)	STATUS_1 (0x99)	STATUS_2 (0x9A)						see SC8721 datasheet
Voltage&Current Monitor	0x30 R	V_BAT LSB (0x30)	V_BAT MSB (0x31)	I_BAT LSB (0x32)	I_BAT MSB (0x33)	V_PWRCAN LSB (0x34)	V_PWRCAN MSB (0x35)	I_PWRCAN LSB (0x36)	I_PWRCAN MSB (0x37)	V_PWR485 LSB (0x38)	V_PWR485 MSB (0x39)	I_PWR485 LSB (0x3A)	I_PWR485 MSB (0x3B)	V_USB LSB (0x3C)	V_USB MSB (0x3D)	I_USB LSB (0x3E)	I_USB MSB (0x3F)	V_BAT: 0~36000mV I_BAT: -8000mA~8000mA V_PWRCAN: 0~36000mV I_PWRCAN: -8000mA~8000mA V_PWR485: 0~36000mV I_PWR485: -8000mA~8000mA
	0x40 R	V_GROVE_RED LSB (0x40)	V_GROVE_RED MSB (0x41)	I_GROVE_RED LSB (0x42)	I_GROVE_RED MSB (0x43)	V_GROVE_BLUE LSB (0x44)	V_GROVE_BLUE MSB (0x45)	I_GROVE_BLUE LSB (0x46)	I_GROVE_BLUE MSB (0x47)									V_USB: 0~36000mV I_USB: -8000mA~8000mA V_GROVE_RED: 0~36000mV I_GROVE_RED: -8000mA~8000mA V_GROVE_BLUE: 0~36000mV I_GROVE_BLUE: -8000mA~8000mA
Detect Status	0x50 R	Charge Status (0x50)	VIN Status (0x51)															Charge Status: 0: no charge 1: charging 2: error VIN Status: 0: no input 1: input present
LED Color	0x60 W/R	USB TypeCB (0x60)	USB TypeCG (0x61)	USB TypeCR (0x62)	Dummy Byte (0x63)	USB TypeAB (0x64)	USB TypeAG (0x65)	USB TypeAR (0x66)	Dummy Byte (0x67)	Grove BlueB (0x68)	Grove BlueG (0x69)	Grove BlueR (0x6A)	Dummy Byte (0x6B)	RS485&CANB (0x6C)	RS485&CANG (0x6D)	RS485&CANR (0x6E)	Dummy Byte (0x6F)	LED Color
	0x70 W/R	Grove RedB (0x70)	Grove RedG (0x71)	Grove RedR (0x72)	Dummy Byte (0x73)	BAT ChargeB (0x74)	BAT ChargeG (0x75)	BAT ChargeR (0x76)	Dummy Byte (0x77)	Power_LB (0x78)	Power_LG (0x79)	Power_LR (0x7A)	Dummy Byte (0x7B)	Power_RB (0x7C)	Power_RG (0x7D)	Power_RR (0x7E)	Dummy Byte (0x7F)	
LED Brightness	0x80 W/R	USB TypeC Brightness (0x80)	USB TypeA Brightness (0x81)	Grove Blue Brightness (0x82)	RS485&CAN Brightness (0x83)	Grove Red Brightness (0x84)	BAT Charge Brightness (0x85)	Power_L Brightness (0x86)	Power_R Brightness (0x87)									Brightness: 0~100
Button	0xA0 R	Button (0xA0)																bit0: 0:button 1 press; 1:button 1 no press
Wakeup Source	0xB0 R	nSTBY_WKUP (0xB0)	VIN_DET (0xB1)	nButton2_WKUP (0xB2)														nSTBY_WKUP: 0,disable; 1,enable(default) VIN_DET: 0,disable; 1,enable(default) nButton2_WKUP: 0,disable; 1,enable(default)
RTC Time	0xC0 W/R	Sec (0xC0)	Minute (0xC1)	Hour (0xC2)	Day (0xC3)	Month (0xC4)	Year (0xC5)	Week Day (0xC6)										Sec: 0~59 Minute: 0~59 Hour: 0~23 Day: 1~31 Month: 1~12 Year: 0~99 Week Day:0x01, Sunday; 02, Monday; 04, Tuesday; 08, Wednesday; 10: Thursday; 20: Friday; 40: Saturday; 80: Sunday
RTC Alarm	0xD0 W/R	Minute (0xD0)	Hour (0xD1)	Day (0xD2)	Alarm Enable (0xD3)													Minute: 0~59 Hour: 0~23 Day: 1~31 Alarm Enable: 1: Alarm Enable; 0: Alarm Disable
System Command	0xE0 W	Power Off (0xE0)	Reset (0xE1)	Download Mode (0xE2)														不用管
Bootloader Version	0xF0 R													Version (0xFC)				Version: bootloader version number
Firmware Version	0xF0 R															Version (0xFE)		Version: firmware version number
I2C Address	0xF0 R																Address (0xFF)	Address: 1~127